Dual-Stack Global Enterprise Architecture

1. Project Objective and Architectural Overview

This project involves the design and deployment of a highly available, dual-stack (IPv4/IPv6) 3-tier enterprise network in Cisco Packet Tracer. The topology simulates a financial services company, requiring strict segregation of duties, gateway redundancy, and secure branch communication over a simulated public internet.
The infrastructure is divided into four logical zones:

- **Zone 1: HQ Campus:** Utilizes a Cisco 3-Tier Campus Architecture (Core, Distribution, Access) focusing on high-speed switching, OSPF internal routing, HSRP, and corporate Wi-Fi.
- **Zone 2: Data Centre / Server Farm:** Hosts centralized IPv4/IPv6 DHCP, DNS, Web, and Syslog services, secured via ACLs.
- **Zone 3: Enterprise Edge & ISP Cloud:** Features dual-homed WAN connectivity utilizing external BGP (eBGP) peering and NAT Overload (PAT).
- **Zone 4: Remote Branch Office:** Securely connected to the HQ Data Centre via an IPsec-over-GRE Virtual Private Network (VPN) tunnel.

2. Hardware Provisioning and Modifications

The project utilizes specific Cisco hardware models, incorporating several engineering pivots to bypass hardware port limitations and simulator constraints.

- **Core Switches:** 2x Catalyst 3650-24PS (CORE-1, CORE-2) provisioned with AC-POWER-SUPPLY modules.
- **Distribution Switches:** 2x Catalyst 3650-24PS (DIST-1, DIST-2) provisioned with AC-POWER-SUPPLY modules.
- **Access Switches:** 4x Catalyst 2960-24TT (ACC-1, ACC-2, ACC-3, ACC-4).
- **Wireless:** 1x WLC-3504 and 2x LAP-PT. The LAP-PT units required physical AC power adapters because the 2960-24TT switches do not support Power over Ethernet (PoE).
- **Data Centre Gateway (Engineering Pivot):** The original ISR 4331 router lacked sufficient built-in Gigabit ports to establish a redundant OSPF connection to both Core switches. The hardware was swapped to a Cisco ISR 2911 router (DC-RTR), which natively features three GigabitEthernet ports, ensuring full High Availability without requiring additional hardware modules.
- **Edge & Branch Routers:** 3x ISR 4331 (EDGE-1, EDGE-2, BRANCH-RTR), each provisioned with a NIM-2T (2-Port Serial) module.
- **ISP Routers (Engineering Pivot):** 2x ISR 4331 (ISP-A, ISP-B). ISP-A required the installation of a second NIM-2T module in slot 0/2 to accommodate three simultaneous serial connections (to EDGE-1, ISP-B, and BRANCH-RTR).

3. Physical Cabling Matrix

The physical topology was wired using the following exact port specifications to ensure symmetrical redundancy.

Core, Distribution, and Access Switching

- **CORE-1 (Gig1/0/1) to CORE-2 (Gig1/0/1):** Copper Cross-Over (L3 EtherChannel).
- **CORE-1 (Gig1/0/2) to CORE-2 (Gig1/0/2):** Copper Cross-Over (L3 EtherChannel).
- **DIST-1 (Gig1/0/1) to DIST-2 (Gig1/0/1):** Copper Cross-Over (L2 EtherChannel).
- **DIST-1 (Gig1/0/2) to DIST-2 (Gig1/0/2):** Copper Cross-Over (L2 EtherChannel).

- **CORE-1 to DIST-1 / DIST-2:** Copper Cross-Over links connecting CORE-1 (Gig1/0/3, Gig1/0/4) to DIST-1 (Gig1/0/3) and DIST-2 (Gig1/0/3).
- **CORE-2 to DIST-1 / DIST-2:** Copper Cross-Over links connecting CORE-2 (Gig1/0/4, Gig1/0/5) to DIST-1 (Gig1/0/4) and DIST-2 (Gig1/0/4).
- **DIST-1 to Access Layer:** Copper Straight-Through links connecting Gig1/0/5, Gig1/0/6, Gig1/0/7, and Gig1/0/8 to Gig0/1 on ACC-1, ACC-2, ACC-3, and ACC-4 respectively.
- **DIST-2 to Access Layer:** Copper Straight-Through redundant links connecting Gig1/0/5, Gig1/0/6, Gig1/0/7, and Gig1/0/8 to Gig0/2 on ACC-1, ACC-2, ACC-3, and ACC-4 respectively.

Data Centre and Edge Routing

- **CORE-1 (Gig1/0/24) to EDGE-1 (Gig0/0/0):** Copper Straight-Through.
- **CORE-2 (Gig1/0/24) to EDGE-2 (Gig0/0/0):** Copper Straight-Through.
- **CORE-1 (Gig1/0/23) to DC-RTR (Gig0/0):** Copper Straight-Through (Primary OSPF Link).
- **CORE-2 (Gig1/0/23) to DC-RTR (Gig0/1):** Copper Straight-Through (Redundant OSPF Link).
- **DC-RTR (Gig0/2) to DC-SW (Gig0/1):** Copper Straight-Through.

WAN, BGP, and Branch Links

- **EDGE-1 (Ser0/1/0) to ISP-A (Ser0/1/0):** Serial DCE.
- **EDGE-2 (Ser0/1/0) to ISP-B (Ser0/1/0):** Serial DCE.
- **ISP-A (Ser0/1/1) to BRANCH-RTR (Ser0/1/0):** Serial DCE.
- **ISP-B (Ser0/1/1) to ISP-A (Ser0/1/2):** Serial DCE.

End-Devices and Wireless Networking

- **WLC-3504 (GigEthernet1) to ACC-4 (Fa0/24):** Copper Straight-Through.
- **LAP-PT 1 (GigEthernet0) to ACC-2 (Fa0/24):** Copper Straight-Through.
- **LAP-PT 2 (GigEthernet0) to ACC-3 (Fa0/24):** Copper Straight-Through.
- **User PCs:** HR-PC connected to ACC-1 (Fa0/1), FIN-PC connected to ACC-2 (Fa0/1), IT-PC connected to ACC-3 (Fa0/1), and MGMT-PC connected to ACC-4 (Fa0/1).

4. Dual-Stack IP Subnetting Scheme

The network was designed with a logically summarized, scalable dual-stack IP schema to support both IPv4 and IPv6 concurrently.

HQ Campus VLANs (Distribution SVI Gateways)

- **VLAN 10 (HR):** 10.1.10.0/24 (GW: .254) | 2001:db8:1:10::/64 (GW: ::1)
- **VLAN 20 (Finance):** 10.1.20.0/24 (GW: .254) | 2001:db8:1:20::/64 (GW: ::1)
- **VLAN 30 (IT):** 10.1.30.0/24 (GW: .254) | 2001:db8:1:30::/64 (GW: ::1)
- **VLAN 50 (Wi-Fi):** 10.1.50.0/24 (GW: .254) | 2001:db8:1:50::/64 (GW: ::1)
- **VLAN 99 (Management):** 10.1.99.0/24 (GW: .254) | 2001:db8:1:99::/64 (GW: ::1)

LAN and Point-to-Point Routing Subnets

- **Data Centre LAN:** 10.2.0.0/24 | 2001:db8:2::/64

- **Remote Branch LAN:** 192.168.10.0/24 | 2001:db8:3::/64

- **CORE-1 to CORE-2:** 10.255.255.16/30 | 2001:db8:A::10/126

- **CORE-1 to DIST-1:** 10.255.255.20/30 | 2001:db8:A::14/126
- **CORE-1 to DIST-2:** 10.255.255.24/30 | 2001:db8:A::18/126
- **CORE-2 to DIST-1:** 10.255.255.28/30 | 2001:db8:A::1C/126
- **CORE-2 to DIST-2:** 10.255.255.32/30 | 2001:db8:A::20/126
- **CORE-1 to EDGE-1:** 10.255.255.0/30 | 2001:db8:A::0/126
- **CORE-2 to EDGE-2:** 10.255.255.4/30 | 2001:db8:A::4/126
- **CORE-1 to DC-RTR:** 10.255.255.8/30 | 2001:db8:A::8/126
- **CORE-2 to DC-RTR:** 10.255.255.12/30 | 2001:db8:A::C/126

Public WAN, GRE, and BGP Subnets

- **EDGE-1 to ISP-A:** 203.0.113.0/30 | 2001:db8:B:1::0/126
- **EDGE-2 to ISP-B:** 198.51.100.0/30 | 2001:db8:B:2::0/126
- **ISP-A to ISP-B:** 8.8.8.0/30 | 2001:db8:B:3::0/126
- **ISP-A to BRANCH-RTR:** 200.1.1.0/30 | 2001:db8:B:4::0/126
- **GRE Tunnel (Edge-1 to Branch):** 172.16.0.0/30 | 2001:db8:C::0/126

5. Implementation Phases & Architecture Pivots

5.1 Layer 2 Switching Foundation

- IPv4 and IPv6 routing capabilities were globally enabled on Core and Distribution switches.
- DIST-1 was configured as the VTP Server for domain FINTECH, synchronizing VLANs 10, 20, 30, 50, and 99 down to all Access layer VTP clients.
- Uplinks between the Access and Distribution layers were configured as 802.1Q Trunks, restricted to native VLAN 99.
- A Layer 2 EtherChannel using LACP (Port-Channel 1) was bundled between DIST-1 and DIST-2 to facilitate HSRP synchronization.
- Access ports facing end-user PCs were assigned to their respective VLANs, and Spanning-Tree PortFast was enabled to bypass the listening/learning phases, ensuring rapid DHCP acquisition.

5.2 Layer 3 Gateways & First-Hop Redundancy (HSRP)

- Switch Virtual Interfaces (SVIs) were generated on DIST-1 and DIST-2 for all VLANs.
- Physical uplinks pointing toward the Core layer on the Distribution switches were converted from switchports to Layer 3 routed ports.
- HSRP Load Balancing was implemented: DIST-1 was configured with increased priority (and preemption) to act as the Active gateway for VLANs 10 and 20. DIST-2 was configured as the Active gateway for VLANs 30

and 50.
- **Engineering Pivot (Simulator Constraint):** Packet Tracer 9.0 lacks the software support for IPv6 First-Hop Redundancy Protocols (HSRP/VRRP). The architecture was adjusted to abandon IPv6 HSRP; the topology relies exclusively on IPv4 HSRP for gateway redundancy demonstration, with static IPv6 SLAAC addresses placed on the SVIs.
- **Engineering Pivot (EtherChannel Order of Operations):** When attempting to assign an IP address to the Core-to-Core EtherChannel, a "Not a convertible port" error occurred. The port-channel was destroyed, the physical ports were explicitly converted to routed ports (no switchport), and the bundle was recreated at Layer 3 to resolve the software bug.

5.3 Internal Dynamic Routing (OSPFv2 and OSPFv3)

- OSPF was activated across the Core, Distribution, Edge, and DC-RTR devices.
- **Engineering Pivot (Dual-Stack OSPF Crash Prevention):** To prevent OSPFv3 from crashing due to the lack of a 32-bit name, both the IPv4 (router ospf 1) and IPv6 (ipv6 router ospf 1) processes were strictly assigned identical, hardcoded Router IDs (e.g., 1.1.1.1 on CORE-1).
- OSPFv2 was enabled via global network wildcard commands, while OSPFv3 was enabled directly at the interface level (ipv6 ospf 1 area 0).
- **Engineering Pivot (Routing Black Hole Fix):** SVI interfaces on the Distribution switches were initially set as passive-interface to prevent Hello packets from reaching end-users. This inadvertently severed the OSPF adjacency between DIST-1 and DIST-2. The configuration was corrected by explicitly removing the passive designation (no passive-interface) on the Native Management VLAN 99, allowing the Distribution switches to form a FULL adjacency across the Layer 2 EtherChannel.

5.4 Centralized Network Services (DHCP & Syslog)

- A dedicated physical DHCP Server was deployed in the Data Centre with separate pools for VLANs 10, 20, 30, and 50.
- DHCP Relay (ip helper-address) was configured on all SVI interfaces at the Distribution layer to intercept client broadcasts and route them to the Data Centre server.
- A centralized Syslog server was activated, with Core, Distribution, and Edge devices configured to forward NOC monitoring logs to its static IP.

5.5 WAN Edge, NAT Overload, and BGP Peering

- External BGP (eBGP) peering was established between EDGE-1 (Private AS 65000) and ISP-A (AS 100), enabling the HQ to learn the simulated internet routes (e.g., Google's 8.8.8.8 DNS).
- NAT Overload (Port Address Translation) was applied to EDGE-1, mapping an Access Control List (ACL) of the internal 10.0.0.0/8 network to the outside Serial interface facing the ISP.
- A static default route (0.0.0.0 0.0.0.0) was injected into the internal OSPF domain using the default-information originate command, successfully allowing internal VLAN PCs to ping public internet addresses.

5.6 Secure Site-to-Site VPN (GRE over IPsec)

- A logical GRE Tunnel (Tunnel 0) was configured between EDGE-1 and BRANCH-RTR, encapsulating private 192.168.10.0/24 branch traffic across the untrusted public serial links.
- The private tunnel subnet (172.16.0.0/30) and Branch LAN were advertised into OSPF Area 0, ensuring the HQ and Branch could route internally without advertising public ISP IP addresses. NAT bypass was achieved by intentionally omitting ip nat outside from the Tunnel interface.
- **Engineering Pivot (Simulator Constraint):** The Cisco 4331 emulator in Packet Tracer does not support

modern IPsec Profiles (crypto ipsec profile). The VPN was rebuilt using legacy Crypto Maps applied directly to the physical Serial WAN interfaces.

- **Engineering Pivot (100% Packet Loss Fix):** Initial ping tests across the tunnel failed due to an asymmetrical cryptography configuration. The ISAKMP policy (AES encryption, SHA hashing, Diffie-Hellman Group 2), Pre-Shared Key, and Transform-Set were missing on BRANCH-RTR. Upon mirroring the Phase 1 and Phase 2 configurations across both routers, the IPsec Security Associations established successfully, allowing the Remote Branch PC to securely ping the Data Centre Web Server.

5.7 Corporate Wireless LAN Integration

- A Cisco WLC-3504 was configured via its GUI Dashboard with a Management IP of 10.1.50.10 residing in VLAN 50.
- A WPA2 Personal WLAN network (FINTECH-CORP) was provisioned on the controller.
- Lightweight Access Points (LAPs) dynamically retrieved DHCP addresses from the Data Centre, located the WLC, and established CAPWAP tunnels to broadcast the corporate SSID.
- **Engineering Pivot (WLC CAPWAP Refresh):** The Packet Tracer initial setup wizard failed to bind the original SSID to the Access Point radios. A secondary WLAN was created to force a CAPWAP configuration refresh, resolving the broadcast issue. Additionally, hardcoded static IPs on testing laptops were removed in favor of DHCP, resolving a simulated L1 helpdesk connectivity ticket and completing the enterprise architecture.